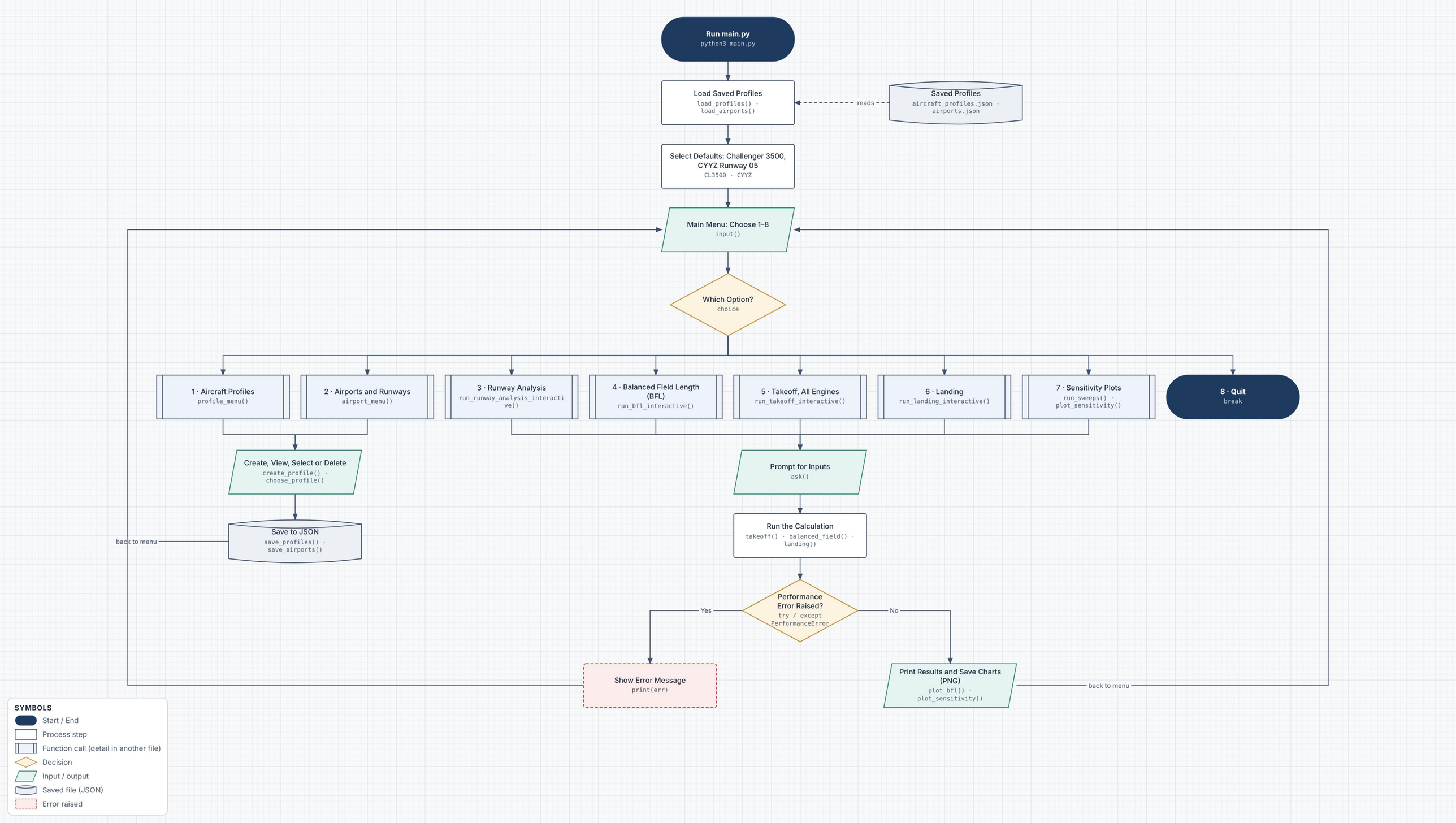
Start Runway Analysis
analyze_runway()
Aircraft Profile
aircraft.py · profiles.py
Airport and Runway
airports.py
Day Conditions: OAT, Wind, Weights, Wet / Dry
ask()
International Standard Atmosphere (ISA)
isa_dev_from_oat() · isa() · sigma()
Headwind and Crosswind, Factored 50% / 150%
wind_components() · factored_headwind()
Build Runway Conditions
make_conditions()
Takeoff Speeds: VS1, VR, V2
stall_speed() · takeoff_speeds()
Ground Roll to VR, Integrated with Runge-Kutta (RK45)
accelerate() · solve_ivp
calls
Forces at Each Step: Thrust, Lift, Drag, Friction
thrust() · lift() · drag()
raises
Rotation Distance
rotation_distance()
Not Enough Thrust or Climb
InsufficientThrustError · ClimbPerformanceError
Transition Arc and Climb to Screen Height
climb_to_screen()
raises
All-Engines Takeoff Distance (TOD)
takeoff()
More Than One Engine?
n_engines > 1
No
Yes
Takeoff Field Length (TOFL) = TOD
Part 23
Balanced Field Length (BFL), Engine Failure at V1
balanced_field() · brentq
Takeoff Field Length (TOFL) = max(BFL, 1.15 × TOD)
Part 25 · §25.113
Landing Distance (LD) and Landing Distance Required (LDR)
landing() · decelerate()
Compare With Declared Distances: TORA, TODA, ASDA, LDA
analyze_runway()
All Margins Positive?
margin ≥ 0
No
Yes
Runway Limited
Takeoff and Landing OK
SYMBOLS
Start / End
Process step
Function call (detail in another file)
Decision
Input / output
Saved file (JSON)
Error raised


Run main.py
python3 main.py
Load Saved Profiles
load_profiles() · load_airports()
reads
Saved Profiles
aircraft_profiles.json · airports.json
Select Defaults: Challenger 3500, CYYZ Runway 05
CL3500 · CYYZ
Main Menu: Choose 1–8
input()
Which Option?
choice
1 · Aircraft Profiles
profile_menu()
2 · Airports and Runways
airport_menu()
3 · Runway Analysis
run_runway_analysis_interactive()
4 · Balanced Field Length (BFL)
run_bfl_interactive()
5 · Takeoff, All Engines
run_takeoff_interactive()
6 · Landing
run_landing_interactive()
7 · Sensitivity Plots
run_sweeps() · plot_sensitivity()
8 · Quit
break
Create, View, Select or Delete
create_profile() · choose_profile()
Save to JSON
save_profiles() · save_airports()
back to menu
Prompt for Inputs
ask()
Run the Calculation
takeoff() · balanced_field() · landing()
Performance Error Raised?
try / except PerformanceError
Yes
No
Show Error Message
print(err)
Print Results and Save Charts (PNG)
plot_bfl() · plot_sensitivity()
back to menu
SYMBOLS
Start / End
Process step
Function call (detail in another file)
Decision
Input / output
Saved file (JSON)
Error raised